MIL-F-36678B
22 March 1976
SUPERSEDING
MIL-F-36678A(DSA-DM)
June 1972 and
MIL-F-36678
July 1970

MILITARY SPECIFICATION

FLASHLIGHT, PATIENT EXAMINING, DISPOSABLE

This specification is approved for use by all the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a disposable, compact, sealed, sanitizable diagnostic flashlight suitable for use in examinations of the oral cavity.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liner, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6230

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1784	Specification for Rigid Poly (Vinyl Chloride) Compounds and Chlorinated Poly (Vinyl Chloride) Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI C18.1	Specifications for Dry Cells and Batteries.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform to the applicable specifications and requirements specified herein. Materials not specifically designated shall be of high commercial quality throughout, and shall be entirely suitable for the intended purpose.

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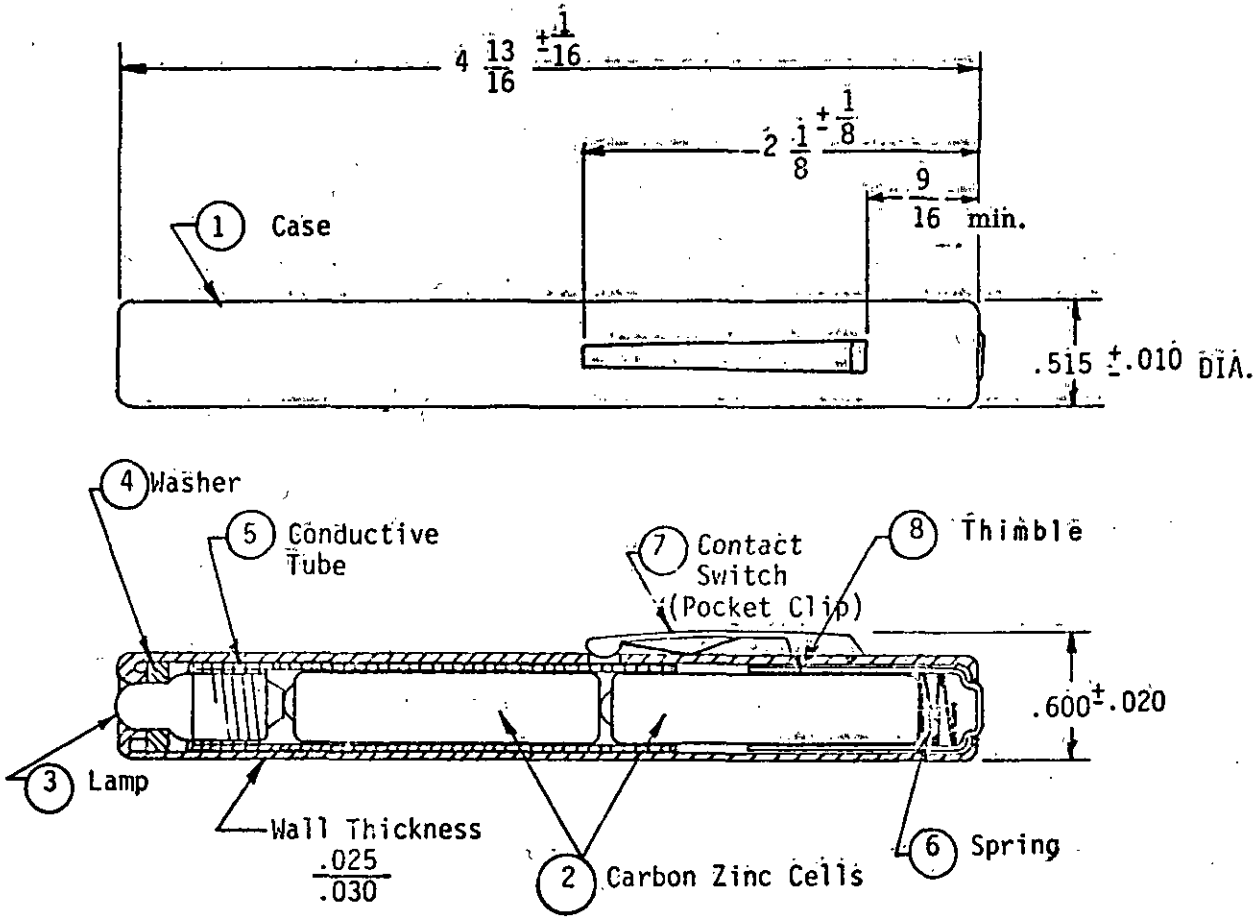
3.2 Style, design, and dimensions. The flashlight shall be a disposable, compact, and diagnostic light suitable for use in examining the dentition of dental ward patients or for performing screening examinations of dental patients in circumstances where lighting fixtures are not available for adequate illumination of the oral cavity. The light shall be permanently sealed and water resistant to permit immersion by cold chemical disinfecting agents such as benzalkonium chloride, or alcohol, when tested as specified in 4.4.1. The light shall consist of a case, two carbon zinc cells, a finely prefocused lamp supported by a ring, a conductive tube, a spring held in an aluminum or other suitable metal thimble, and a contact switch which also serves as a pocket clip. The style, design, and dimensions shall be as shown in figure 1. Each individual flashlight shall be placed in a cardboard sleeve with a cutout for fitting sleeve under clip. Sleeve shall be designed to prevent clip from accidentally contacting the conductive tube during shipment or storage.

3.2.1 Case. The case shall be fabricated of extruded, rigid, white polyvinyl chloride in accordance with Class 1433-D of American Society for Testing and Materiels (ASTM) Standard D 1784. The case shall be rolled inward and downward on the distal end of the light to form a compression seal with the lamp and the lamp support ring. The case shall be rolled inward over the thimble on the proximal end.

3.2.2 Battery. The battery shall consist of two Leclanche cylindrical dry cells (carbon zinc), each having a nominal open circuit voltage of 1.5 volts or 3.0 volts when connected in series. The battery cells shall be American National Standards Institute (ANSI) designation AAA general purpose flashlight cell, and shall conform to applicable paragraphs of ANSI Standard C18.1, except that the delayed 40 minute intermittent test shall be run 12 months from date of manufacture in lieu of 6 months. The terminals shall be flat, contact-type terminals, and the cells may be unjacketed.

3.2. Incandescent lamp. The incandescent lamp shall be recessed insofar as possible within the distal end of the case to preclude breakage and shall be supported by a plastic, translucent, light-diffusing ring. The lamp shall be a clear, finely prefocused bulb with a miniature screw base, number 222, with TL-3 bulb design and C-6 filament. The lamp shall be rated 2.25 volts and 0.25 amperes $\pm$ 10 percent. The overall length of the lamp shall be 15/16 inch maximum.

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All dimensions are in inches.

FIGURE 1

FLASHLIGHT

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3.2.4 Conductive tube. The conductive tube shall be fabricated of 0.006 inch electrical grade insulating paper with a 0.001 inch brass foil laminated to the outside of the paper. Both the paper and the brass shall be spiral wrapped to form the tube. The distal end of the tube shall be folded inward on itself to make positive electrical contact with the screw base of the bulb. The proximal end of the tube shall extend beyond the cutout in the case for the contact switch.

3.2.5 Contact switch. The contact switch shall be fabricated of nickel-plated, medium carbon steel. The contact switch shall be in contact with the thimble, and shall complete the electrical circuit when depressed to touch conductive tube. The switch shall also serve as a pocket clip. The contact switch shall be securely attached to the outer case.

3.2.6 Spring. The spring shall be fabricated of 0.035 ± 0.001 inch steel music wire, and shall be inserted inside the thimble to provide necessary tension for holding components in place.

3.3 Bulb centering. The light beam shall fall within a 3-1/2 inch diameter circle, when tested as specified in 4.4.2.

3.4 Beam divergence. The beam divergence shall be no less than 10 degrees nor more than 45 degrees, when tested as specified in 4.4.3.

3.5 Weight. The flashlight shall weigh 24.0 ± 1.0 grams.

3.6 Finish. The flashlight shall have a white satin finish, except that the contact switch shall have a bright finish, and the glass of the lamp shall be clear.

3.7 Delivery schedule. No more than two months shall have elapsed from date of manufacture of the battery cells to date of shipment to the Government.

3.8 Quantity. Twelve flashlights, each in an individual sleeve, shall constitute a unit of issue.

3.9 Marking. Each flashlight shall be legibly and permanently marked on the clip with the manufacturer's name or registered trademark.

3.10 Shelf life. The flashlight shall comply with the specification requirements for a minimum of 12 months from date of delivery. Inspection test date shall be 12 months from the date of delivery.

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3.11 Workmanship. Workmanship shall be first class throughout. Flashlights shall be free from defects which either detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in that specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the polyvinyl chloride case and the battery, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I, II, and III (see 4.2.1, 4.2.2 and 4.3) and 4.3.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one flashlight.

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Table I. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one flashlight. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Tests

Characteristic	Requirement	Test procedure	Inspection level
Immersion test	3.2	4.4.1	S-2
Capacity test	3.2.2	ANSI Std. C18.1	1/
Bulb centering	3.3	4.4.2	S-2
Beam divergence	3.4	4.4.3	S-2
Weight	3.5	S.M.I. 2/	S-2

1/ A minimum of 6 dry cell samples per batch number shall be used for the initial capacity test, and a minimum of 6 dry cell samples shall be stored at 70° ± 2°F. by the contractor or battery manufacturer for the 12 month delayed capacity test.

2/ SMI - Standard Measuring Instrument.

4.2.3 Raw material. A sufficient quantity of each lot of polyvinyl chloride shall be selected for testing.

4.3 Examination. The flashlight shall be examined for defects including, but not limited to, those listed in table III.

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Table III. Classification of defects

Category	Defects
Major	
101	Style and design not as shown on figure (in-process inspection).
102	Flashlight not constructed as specified (in-process inspection).
103	Contact switch fails to turn on light when pressed and turn off light when released.
104	Cardboard sleeve not as specified.
105	Burrs or sharp edges present.
Minor A	
201	Scratches or dents present.
202	Finish not as specified.
203	Color of case not as specified.
204	Identification marking not as specified.
Minor B	
221	Dimensions not as specified.

4.3.1 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

4.4.1 Immersion test. The flashlight shall be completely immersed for 30 minutes at $70^{\circ} \pm 5^{\circ}$F. in benzalkonium chloride solution (1:1,000). The flashlight shall be wiped dry and immediately turned on and off to test for proper operation.

4.4.2 Bulb centering. The flashlight shall be mounted perpendicular to, and four inches from, a flat surface. The target circle shall be centered at the point where the axis of the flashlight intersects the plane.

4.4.3 Beam divergence test. The maximum angle of beam divergence shall be measured with a protractor when the beam is played across a flat dark surface.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified, (see 6.1).

5.1.1 Unit of issue. One package containing twelve flashlights, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each light shall be inserted into a snug fitting paperboard tube. The tube shall be between the light and the pocket clip to prevent the light from being accidentally activated. Twelve lights shall be further packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Twelve unit packages shall be packed in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A or C, as specified (see 6.1).

5.2.1 Level A. Four intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 304. of PPP-B-636.

5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. Type II Shelf Life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE BETWEEN 2° - 8°C. (35° - 46°F.)".

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Type II Shelf Life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE BETWEEN 2° - 8°C. (35° - 46°F.)".

NOTE: The unrefrigerated shipping time for this item shall be not more than thirty (30) days (720 hours).

DD Form 1502, 1502-1 or 1502-2 as specified in MIL-STD-129, shall not be required.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog: 6230-00-125-5528, FLASHLIGHT, Patient Examining, Disposable, 12s.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6230-0227

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-36678B FLASHLIGHT, PATIENT EXAMINING, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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